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Substrate process apparatus and substrate process method using the same

Abstract

A substrate process apparatus includes a chamber body providing a process space and a chamber lid coupled to the chamber body, the chamber lid includes a focus adapter providing a diffusion space of which width increases in a direction approaching a bottom of the focus adapter, a lid body surrounding the focus adapter, and a coupling adapter supporting the focus adapter, the lid body provides a coupling through hole recessed downwardly from an upper surface of the lid body, the coupling adapter provides a coupling recess recessed downwardly from an upper surface of the coupling adapter, and the coupling recess is spaced apart outwardly from the diffusion space and is located below the coupling through hole.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7658802	12/2009	Fu et al.	N/A	N/A
2013/0240142	12/2012	Foo et al.	N/A	N/A
2017/0301514	12/2016	Cho	N/A	H01J 37/32449
2018/0230624	12/2017	Dube et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
100542031	12/2005	KR	N/A
100683416	12/2006	KR	N/A
1020070045536	12/2006	KR	N/A
100801857	12/2007	KR	N/A
101454861	12/2013	KR	N/A
101724100	12/2016	KR	N/A
1020190108176	12/2018	KR	N/A
102121598	12/2019	KR	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This U.S. non-provisional patent application claims priority under 35 U.S.C.§ 119 to Korean Patent Application No. 10-2022-0087604, filed on Jul. 15, 2022, in the Korean Intellectual Property Office, the entire contents of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

(2) The inventive concept relates to a substrate process apparatus and a substrate process method using the same, and more particularly, relates to a substrate process apparatus capable of preventing damage to a coupling screw and a substrate process method using the same.

(3) A semiconductor device may be manufactured through various processes. For example, the semiconductor device may be manufactured through a photo process, an etching process, a deposition process, and the like on a wafer such as silicon. A photoresist may be applied on a wafer in the photo process. The photoresist material on the wafer should be removed after performing certain processes. To remove the photoresist material from the wafer, an ashing process may be used. In the ashing process, plasma may be used. The plasma may remove the photoresist from the substrate.

SUMMARY

(4) An object of the inventive concept is to provide a substrate process apparatus capable of preventing damage to a coupling screw and a substrate process method using the same.

(5) An object of the inventive concept is to provide a substrate process apparatus capable of improving yield by preventing damage/defects to a substrate and a substrate process method using the same.

(6) An object of the inventive concept is to provide a substrate process apparatus capable of being easily and quickly assembled and a substrate process method using the same.

(7) An object of the inventive concept is to provide a substrate process apparatus capable of effectively maintaining a vacuum in a process space and a substrate process method using the same.

(8) The problem to be solved by the inventive concept is not limited to the problems mentioned above, and other problems not mentioned will be clearly understood by those skilled in the art from the following description.

(9) A substrate process apparatus according to some embodiments of the inventive concept may include a chamber body providing a process space, and a chamber lid coupled to the chamber body, the chamber lid may include a focus adapter providing a diffusion space of which width increases in a direction approaching a bottom of the focus adapter, a lid body surrounding the focus adapter, and a coupling adapter supporting the focus adapter, the lid body may provide a coupling through hole recessed downwardly from an upper surface of the lid body, the coupling adapter may provide a coupling recess recessed downwardly from an upper surface of the coupling adapter, and the coupling recess may be spaced apart outwardly from the diffusion space and is located below the coupling through hole.

(10) A substrate process apparatus according to some embodiments of the inventive concept may include a chamber body defining a process space configured such that a substrate is disposed in the process space, and a chamber lid on the chamber body, the chamber lid may include a focus adapter defining a diffusion space connected to the process space, a lid body surrounding the focus adapter, and a coupling adapter fixing the focus adapter to the lid body, the coupling adapter may include a support ring and a plurality of support blocks protruding inwardly from the support ring, the plurality of support blocks may be spaced apart from each other in a circumferential direction along the support ring, and the focus adapter may be positioned on the plurality of support blocks.

(11) A substrate process method according to some embodiments of the inventive concept may include placing a substrate in a substrate process apparatus, supplying a gas to the substrate process apparatus, generating plasma using the gas supplied to the substrate process apparatus, and processing the substrate with the plasma, the substrate process apparatus may include a chamber body providing a process space in which the substrate is disposed and a chamber lid on the chamber body, the chamber lid may include a focus adapter providing a diffusion space in which the plasma is diffused, a lid body surrounding the focus adapter, and a coupling adapter supporting the focus adapter, the lid body may provide a coupling through hole connected to an upper surface of the lid body, and the coupling adapter may provide a coupling recess positioned below the coupling through hole.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Example embodiments will be more clearly understood from the following description taken in conjunction with the accompanying drawings. The accompanying drawings represent non-limiting, example embodiments as described herein.

(2) FIG. **1** is a cross-sectional view illustrating a substrate process apparatus according to embodiments of the inventive concept.

(3) FIG. **2** is an exploded cross-sectional view illustrating a substrate process apparatus according to embodiments of the inventive concept.

(4) FIG. **3** is an enlarged cross-sectional view of region "X" of FIG. **1**.

(5) FIG. **4** is an exploded cross-sectional view of FIG. **3**.

(6) FIG. **5** is a plan view taken along line I-I′ of FIG. **1**.

(7) FIG. **6** is a plan view illustrating an enlarged region "Y" of FIG. **5**.

(8) FIG. **7** is a flowchart illustrating a substrate process method according to embodiments of the inventive concept.

(9) FIGS. **8** to **10** are cross-sectional views sequentially illustrating a substrate process method according to the flowchart of FIG. **7**.

(10) FIG. **11** is a plan view taken along line I-I′ of FIG. **1**.

(11) FIG. **12** is a plan view illustrating an enlarged region "Z" of FIG. **11**.

DETAILED DESCRIPTION

(12) Hereinafter, embodiments of the inventive concept will be described with reference to the accompanying drawings. Like reference numerals may refer to like elements throughout the specification.

(13) FIG. **1** is a cross-sectional view illustrating a substrate process apparatus according to embodiments of the inventive concept. FIG. **2** is an exploded cross-sectional view illustrating a substrate process apparatus according to embodiments of the inventive concept.

(14) Hereinafter, D**1** may be referred to as a first direction, D**2** intersecting the first direction D**1** may be referred to as a second direction, and D**3** intersecting each of the first direction D**1** and the second direction D**2** may be referred to as a third direction. The first direction D**1** may be a vertical direction. In addition, each of the second direction D**2** and the third direction D**3** may be a horizontal direction. For example, the first, second and third directions D**1**, D**2** and D**3** may be perpendicular to each other.

(15) Referring to FIGS. **1** and **2**, a substrate process apparatus may be provided. The substrate process apparatus may be an apparatus for processing a substrate using plasma. The term "substrate" used herein may denote a silicon (Si) wafer, but is not limited thereto. For example, the term "substrate" may denote a substrate itself, or a stack structure including a substrate and predetermined layers or films formed on a surface of the substrate. In addition, the term "surface of a substrate" may denote an exposed surface of the substrate itself, or an external surface of a predetermined layer or a film formed on the substrate. The substrate process apparatus may be, for example, an apparatus for an ashing process. For example, the substrate process apparatus may be an ashing chamber. A photoresist on the substrate may be removed by the substrate process apparatus. For example, the photoresist on the substrate disposed in the substrate process apparatus may be removed from the substrate by plasma. However, the inventive concept is not limited thereto, and the substrate process apparatus may be another type of chamber/apparatus for an etching process and/or a deposition process. Hereinafter, for convenience, the chamber/apparatus is illustrated and described based on the ashing chamber.

(16) The substrate process apparatus may include a chamber body **1**, a chamber lid **3**, a baffle **5**, a chuck **7**, a plasma generator **9**, a gas supplier GS, a discharge pump VP, and the like.

(17) The chamber body **1** may provide a process space **11***h*. The process space **11***h* provided by the chamber body **1** may have an open top. For example, when the chamber lid **3** is separated from the chamber body **1**, the process space **11***h* may be exposed to the outside. When the chamber lid **3** is coupled to the chamber body **1**, the process space **11***h* may be separated from the outside. A substrate may be disposed in the process space **11***h*. When the substrate is disposed in the process space **11***h*, a process for the substrate may be performed. For example, when the substrate process apparatus is an ashing chamber, an ashing process may be performed on the substrate when the substrate is in the process space **11***h*. The chamber body **1** may further provide an outlet **13***h*. The outlet **13***h* may be connected to the process space **11***h*. The outlet **13***h* may be disposed below the process space **11***h*, but is not limited thereto. The outlet **13***h* may be connected to the discharge pump VP. A fluid in the process space **11***h* may be discharged from the chamber and/or the process space **11** through the outlet **13***h* by a vacuum pressure provided by the discharge pump VP.
(18) The chamber lid **3** may be coupled on the chamber body **1**. The chamber lid **3** may be detachably coupled to the chamber body **1**. The chamber lid **3** may selectively close the process space **11***h*. The chamber lid **3** may include a focus adapter **33**, a lid body **31**, a coupling adapter **35**, a coupling screw **37**, and the like.
(19) The focus adapter **33** may provide a diffusion space **33***h*. For example, the diffusion space **33***h* may be defined by an inner surface of the focus adapter **33**. The diffusion space **33***h* may become wider downwardly. For example, horizontal area/width of the diffusion space **33***h* may increase in a direction approaching a bottom of the focus adapter **33**. For example, a diameter of the diffusion space **33***h* may linearly increase downwardly. For example, diameters/widths of the diffusion space **33***h* in a horizontal direction may increase in a direction moving downwards. The diffusion space **33***h* may be connected to the plasma generator **9**. For example, the diffusion space **33***h* may be connected to a plasma generating space **9***h* provided by the plasma generator **9**. In addition, the diffusion space **33***h* may be connected to the process space **11***h*. For example, the diffusion space **33***h* may be connected to the process space **11***h* through the baffle **5**. Plasma produced by the plasma generator **9** may be horizontally and downwardly diffused through the diffusion space **33***h* and the baffle **5**. The plasma passing through the baffle **5** may be introduced into the process space **11***h*. Details thereon will be described later.
(20) The lid body **31** may surround the focus adapter **33**. For example, the lid body **31** may surround the focus adapter **33** in a horizontal direction. For example, the lid body **31** may horizontally overlap the whole focus adapter **33**. An inner surface of the lid body **31** may be spaced apart from an outer surface of the focus adapter **33** outwardly. A term “outwardly” as used herein may mean a direction away from a vertical axis of the chuck **7**. For example, the outward direction may be a direction receding from an axis of symmetry passing through the center of the chuck **7** and extending in a vertical direction. The lid body **31** may be selectively coupled to the chamber body **1**. The lid body **31** may include or be formed of, for example, quartz, but is not limited thereto.
(21) Spatially relative terms, such as “horizontal,” “vertical,” “beneath,” “below,” “lower,” “above,” “upper,” “top,” “bottom” and the like, may be used herein for ease of description to describe positional relationships. It will be understood that the spatially relative terms encompass different orientations of the device in addition to the orientation depicted in the figures.
(22) The coupling adapter **35** may support the focus adapter **33**. The coupling adapter **35** may fix the focus adapter **33** to the lid body **31**. The coupling adapter **35** may be fixed to the lid body **31**.
(23) The coupling screw **37** may couple the coupling adapter **35** and the lid body **31** to each other. For example, the coupling adapter **35** may be fixed to the lid body **31** by the coupling screw **37**. The coupling screw **37** may be exposed on an upper surface of the lid body **31**. The coupling screw **37** may not be exposed to the process space **11***h*. Details thereon will be described later.
(24) The baffle **5** may be positioned between the chamber body **1** and the chamber lid **3**. The baffle **5** may be located above the process space **11***h* and below the diffusion space **33***h*. The baffle **5** may

provide an inlet **5***h* (refer to FIG. **3**). The inlet **5***h* may allow the fluid in the diffusion space **33***h* to move into the process space **11***h*. As shown in FIG. **2**, the baffle **5** may be fixed to the focus adapter **33**, the coupling adapter **35**, and/or the lid body **31**. However, the inventive concept is not limited thereto, and the baffle **5** may be coupled to the chamber body **1**.

(25) The chuck **7** may be located in the chamber body **1**. For example, the chuck **7** may be placed in the process space **11***h* provided by the chamber body **1** and may be supported by the chamber body **1**. The chuck **7** may support the substrate in the process space **11***h*. The chuck **7** may fix the substrate at a predetermined position. To this end, the chuck **7** may include or may be a vacuum chuck, an electrostatic chuck (ESC), or the like.

(26) The plasma generator **9** may be located on the chamber lid **3**. The plasma generator **9** may form an electric field and/or a magnetic field to convert some of gas into plasma. The plasma generator **9** may be, for example, an induced coupled plasma (ICP) generator. In this case, the plasma generator **9** may include a coil **91** and a plasma generating body **93**. The plasma generating body **93** may provide a plasma generating space **9***h*. The plasma generating space **9***h* may be connected to the diffusion space **33***h*. The coil **91** may surround the plasma generating body **93**. However, the inventive concept is not limited thereto, and the plasma generator **9** may generate plasma in other ways.

(27) The gas supplier GS may supply gas to the plasma generator **9**. To this end, the gas supplier GS may include a gas tank, a compressor, and/or a valve.

(28) The discharge pump VP may be connected to the outlet **13***h*. The discharge pump VP may be, for example, a vacuum pump. By the discharge pump VP, the process space **11***h* may be maintained in a substantially vacuum state, e.g., in a vacuum state.

(29) FIG. **3** is an enlarged cross-sectional view of region “X” of FIG. **1**, and FIG. **4** is an exploded cross-sectional view of FIG. **3**.

(30) Referring to FIGS. **3** and **4**, the lid body **31** may provide a coupling through hole **31***h*. The coupling through hole **31***h* may be formed to be depressed from an upper surface of the lid body **31** downwardly. For example, the coupling through hole **31***h* may vertically extend from an upper surface of the lid body **31** downwardly. Although not shown, a screw structure may be provided on an inner surface defining the coupling through hole **31***h*. The coupling screw **37** may be inserted into the coupling through hole **31***h*. The lid body **31** may include an upper body **311** and a lower body **313**. An outer diameter and/or inner diameter of the upper body **311** may increase downwardly. For example, the outer/inner diameters/widths of the upper body **311** in a horizontal direction may increase in a direction moving downwards. The lower body **313** may be coupled to a lower end of the upper body **311**, e.g., at its upper end. The lower body **313** may be coupled to the chamber body **1**, e.g., at its lower end. The lower body **313** and the upper body **311** may be integrally formed, e.g., as one body without boundaries between them, but the inventive concept is not limited thereto. The coupling through hole **31***h* may be provided in the lower body **313**. The coupling through hole **31***h* may be connected to an upper surface **313***u* of the lower body **313**. For example, the coupling through hole **31***h* may be formed from the upper surface **313***u* of the lower body **313** to a bottom surface of the lower body **313**. For example, the coupling through hole **31***h* may penetrate the lower body **313** vertically.

(31) The focus adapter **33** may be supported by the coupling adapter **35**. The focus adapter **33** may include an upper adapter **331**, a lower adapter **333**, and a support member **335**.

(32) An outer diameter and/or inner diameter of the upper adapter **331** may increase downwardly. For example, the outer/inner diameters/widths of the upper adapter **331** in a horizontal direction may increase in a direction moving downwards. The upper adapter **331** may be disposed to an inner side of the upper body **311**. An outer surface of the upper adapter **331** may be spaced apart from an inner surface of the upper body **311** inwardly. Accordingly, a gap “G” may be formed between the upper adapter **331** and the upper body **311**.

(33) The lower adapter **333** may be coupled to and disposed under the upper adapter **331**. The

lower adapter **333** may be integrally formed with the upper adapter **331**, e.g., as one body without boundaries between them, but is not limited thereto. The lower adapter **333** may be located inside the lower body **313**. An outer surface of the lower adapter **333** may be spaced apart from an inner surface of the lower body **313** inwardly. Accordingly, the gap "G" may extend between the lower adapter **333** and the lower body **313**.

(34) The support member **335** may be coupled to the lower adapter **333**. The support member **335** may protrude outwardly from one side of the outer surface of the lower adapter **333**. The support member **335** may be supported by the coupling adapter **35**. For example, the support member **335** may be located on one side of the coupling adapter **35**. For example the support member **335** may contact an upper surface of the coupling adapter **35**. The term "contact," "contacting," "contacts," or "in contact with," as used herein, refers to a direct connection (i.e., touching) unless the context clearly indicates otherwise.

(35) The coupling adapter **35** may provide a coupling recess **35***h***1**. The coupling recess **35***h***1** may be recessed downwardly from an upper surface **35***u* of the coupling adapter **35**. The coupling recess **35***h***1** may be located outside the diffusion space **33***h*. The coupling recess **35***h***1** may be located below the coupling through hole **31***h*. The coupling recess **35***h***1** may be connected to the coupling through hole **31***h*. Although not shown, a screw structure may be provided on an inner surface defining the coupling recess **35***hl*. The coupling screw **37** may be inserted into the coupling recess **35***hl*.

(36) The coupling adapter **35** may provide a sealing hole. The sealing hole may be recessed downwardly from the upper surface **35***u* of the coupling adapter **35**. The sealing hole may extend in a circumferential direction. A plurality of sealing holes may be provided in the coupling adapter **35**. For example, a first sealing hole **35***h***2** and a second sealing hole **35***h***3** may be provided in the coupling adapter **35**. The first sealing hole **35***h***2** may be located inside the coupling recess **35***hl*. The second sealing hole **35***h***3** may be located outside the coupling recess **35***hl*.

(37) The coupling adapter **35** may include a support ring **351** and a support block **353**.

(38) The support ring **351** may have a ring shape extending in a circumferential direction. The support ring **351** may be provided with the coupling recess **35***h***1** in the support ring **351**. For example, the coupling recess **35***h***1** may be connected to an upper surface of the support ring **351**. A height/depth of the coupling recess **35***h***1** may be smaller than a thickness of the support ring **351**. Accordingly, the coupling recess **35***h***1** may be spaced upwardly from a lower surface of the coupling adapter **35**. For example, a bottom surface of the coupling recess **35***h***1** may be at a higher level than a bottom surface of the coupling adapter **35**. Therefore, the coupling recess **35***h***1** may not be connected to a lower space of the support ring **351**. For example, the coupling recess **35***h***1** may not be exposed to the process space **11***h*. The support ring **351** may include or be formed of, but is not limited to, aluminum (Al).

(39) The support block **353** may be coupled to the support ring **351**. The support block **353** may protrude inwardly from the support ring **351**. The support block **353** may be integrally formed with the support ring **351**, e.g., as one body without boundaries between the support block **353** and the support ring **351**, but is not limited thereto. The support block **353** may support the focus adapter **33**. For example, the support member **335** may be positioned on the support block **353** and supported by the support block **353**. For example, the support member **335** may contact the support block **353**. The support block **353** may include or be formed of aluminum (Al), but is not limited thereto. A plurality of support blocks **353** may be provided in the coupling adapter **35**. The plurality of support blocks **353** may be disposed to be spaced apart from each other in a circumferential direction along the support ring **351**. However, hereinafter, unless otherwise specified, the support block **353** will be described in the singular. For example, the single support block **353** described below may represent one of the plural support blocks **353** or may describe embodiments including a single support block **353** in the coupling adapter **35**. For example, when the coupling adapter **35** includes a single support block **353**, the single support block **353** may extend in a circumferential

direction along the support ring **351** to form a ring shape or may extend in the circumferential direction along the support ring **351** more than a half of the support ring **351**. The details thereof will be described later with reference to FIGS. **5** and **6**.

(40) The chamber lid **3** (refer to FIG. **1**) may further include a sealing ring **39**. The sealing ring **39** may be, for example, an O-ring. The sealing ring **39** may be located on the support ring **351**. The sealing ring **39** may extend in a circumferential direction along the support ring **351**. The chamber lid **3** may include a plurality of sealing rings **39**. For example, a first sealing ring **391** and a second sealing ring **393** may be provided in the chamber lid **3**. The first sealing ring **391** may be located inside the coupling recess **35***h***1** in a plan view. The second sealing ring **393** may be located outside the coupling recess **35***h***1** in the plan view. The first sealing ring **391** may be inserted into the first sealing hole **35***h***2**. The second sealing ring **393** may be inserted into the second sealing hole **35***h***3**. By the first sealing ring **391** and the second sealing ring **393**, fluid may be prevented from leaking into a gap between the lid body **31** and/or the coupling adapter **35** and the coupling screw **37**.

(41) FIG. **5** is a plan view taken along line I-I′ of FIG. **1**, and FIG. **6** is a plan view illustrating an enlarged region “Y” of FIG. **5**.

(42) Referring to FIGS. **5** and **6**, the coupling recess **35***h***1** may be provided in plurality. A plurality of coupling recesses **35***h***1** may be spaced apart from each other in a circumferential direction. However, hereinafter, for convenience, the coupling recess **35***h***1** will be described in the singular. For example, the coupling recess **35***h***1** may represent each of the plurality of the coupling recesses **35***h***1**.

(43) The support ring **351** may extend in a circumferential direction. Each of the first sealing ring **35***h***2** and the second sealing ring **35***h***3** may extend in a circumferential direction. The plurality of support blocks **353** may be disposed to be spaced apart from each other in a circumferential direction along the support ring **351**.

(44) FIG. **7** is a flowchart illustrating a substrate process method according to embodiments of the inventive concept.

(45) Referring to FIG. **7**, a substrate process method “S” may be provided. The substrate process method “S” may be a method of performing a process on a substrate using the substrate process apparatus described with reference to FIGS. **1** to **6**. The substrate process method “S” may include placing a substrate in the substrate process apparatus in S**1**, supplying a gas into the substrate process apparatus in S**2**, generating plasma in S**3**, and processing the substrate in S**4**.

(46) Hereinafter, the substrate process method “S” of FIG. **7** will be sequentially described with reference to FIGS. **8** to **10**.

(47) FIGS. **8** to **10** are cross-sectional views sequentially illustrating a substrate process method according to the flowchart of FIG. **7**.

(48) Referring to FIGS. **8** and **7**, the placing of the substrate in the substrate process apparatus in S**1** may include disposing a substrate “W” on a chuck **7**. The substrate “W” may be a silicon (Si) wafer, but is not limited thereto. In embodiments, the substrate “W” may be in a state in which a photoresist is applied. For example, after the photoresist is applied on the substrate “W”, the substrate “W” in a state in which a pattern is formed through an etching process may be disposed on the chuck **7**. However, the inventive concept is not limited thereto, and the substrate “W” may be in another state. The chuck **7** may fix the substrate “W”. For example, the chuck **7** may use a vacuum pressure and/or an electrostatic force to fix the substrate “W” at a certain position.

(49) Referring to FIGS. **9** and **7**, the supplying of the gas into the substrate process apparatus in S**2** may include supplying the gas “G” by a gas supplier GS. The gas “G” may be supplied to a plasma generator **9**. The gas “G” may be introduced into a plasma generating space **9***h*.

(50) The generating of the plasma in S**3** may include converting at least a portion of the gas in the plasma generating space **9***h* into plasma PL by the plasma generator **9**. For example, at least a portion of the gas may be converted into the plasma PL by an electric field and/or a magnetic field formed by a coil **91**.

(51) Referring to FIGS. **9**, **10** and **7**, the processing of the substrate in S**4** may include moving the plasma PL onto the substrate “W”. For example, the plasma PL may pass through a diffusion space **33***h* and a baffle **5** and may be diffused in a horizontal direction, and then may move onto the substrate “W”. When the substrate process method “S” is a method of performing an ashing process, the photoresist on the substrate “W” may be removed by the plasma PL. However, the inventive concept is not limited thereto, and an etching process and/or a deposition process by plasma PL may be performed.
(52) Referring to FIG. **10**, some plasma PLL may move through a gap “G”. Some plasma PLL moving through the gap “G” may be prevented from reaching the coupling screw **37** by a first sealing ring **391** and a second sealing ring **393**.
(53) According to the substrate process apparatus and the substrate process method using the same according to embodiments of the inventive concept, the plasma may be prevented from reaching the coupling screw. For example, the plasma may be prevented from reaching the coupling screw by the sealing ring. For example, the plasma may not reach the coupling screw. Accordingly, damage to the coupling screw may be prevented. For example, the coupling screw may be inserted downwardly from the upper surface of the lid body, and thus the coupling screw may be prevented from being exposed to the process space. Accordingly, damage to the coupling screw may be prevented to increase lifespan of the coupling screw. In addition, particles generated by damage to the coupling screw may be prevented. Accordingly, the substrate may be prevented from being contaminated by particles or the like. The damage to the substrate may be prevented, and thus manufacturing yield may be improved. For example, the aforementioned structures of the embodiments may be beneficial to the lifespans of the substrate processing apparatus and its parts, and beneficial to the yield of the substrates processed by the apparatus.
(54) According to the substrate process apparatus and the substrate process method using the same according to embodiments of the inventive concept, the coupling adapter may be provided in a form of an integrated ring. Accordingly, the chamber lid may be combined to the chamber body **1** fast and simple.
(55) According to the substrate process apparatus and the substrate process method using the same according to embodiments of the inventive concept, the coupling screw may be inserted from the upper surface of the lid body downwardly, and thus the assembly of the coupling screw may be simple and quick.
(56) According to the substrate process apparatus and the substrate process method using the same according to embodiments of the inventive concept, the coupling screw may be surrounded by the sealing ring. Therefore, even when a part of the coupling screw is exposed to the external space above the chamber lid, the external space and the process space may be prevented from being connected through the gap between the coupling screw and the chamber lid. Therefore, it is possible to maintain a vacuum state of the process space.
(57) FIG. **11** is a plan view taken along line I-I′ of FIG. **1**, and FIG. **12** is a plan view illustrating an enlarged region “Z” of FIG. **11**.
(58) Hereinafter, descriptions of contents the same or substantially the same as or similar to those described with reference to FIGS. **1** to **10** may be omitted. For example, omitted descriptions of elements may be the same as the descriptions of corresponding elements of embodiments described above.
(59) Referring to FIG. **11**, a coupling adapter **35***a* may be provided. However, unlike the coupling adapter **35** described with reference to FIGS. **5** and **6**, the coupling adapter **35***a* of FIGS. **11** and **12** may not have a ring shape. For example, a plurality of discrete/separate coupling adapters **35***a* may be provided. For example, eight coupling adapters **35***a* may be provided. However, the inventive concept is not limited thereto, and the number of coupling adapters **35***a* may vary depending on specific design application. The plurality of coupling adapters **35***a* may be disposed to be spaced apart from each other in a circumferential direction. However, hereinafter, for convenience, the

coupling adapter **35***a* will be described in the singular. For example, the coupling adapter **35***a* described below may represent each of the plural coupling adapters **35***a*.
(60) Referring to FIG. **12**, the coupling adapter **35***a* may provide a sealing hole **35***ah***2**. The sealing hole **35***ah***2** may be recessed downwardly from an upper surface of the coupling adapter **35***a*. The sealing hole **35***ah***2** may surround the coupling recess **35***ah***1**, e.g., in a plan view.
(61) According to the substrate process apparatus and the substrate process method using the same of the inventive concept, the damage to the coupling screw may be prevented.
(62) According to the substrate process apparatus and the substrate process method using the same of the inventive concept, the damage to the substrate may be prevented and the yield may be improved. For example, the aforementioned structures of the embodiments of FIGS. **11** and **12** may be beneficial to the lifespans of the substrate processing apparatus and its parts, and beneficial to the yield of the substrates processed by the apparatus.
(63) According to the substrate process apparatus and the substrate process method using the same of the inventive concept, the assembly may be simple and quick.
(64) According to the substrate process apparatus and the substrate process method using the same of the inventive concept, the vacuum in the process space may be maintained. For example, the substrate process apparatus of the above embodiments may be beneficial to maintaining vacuum states in the diffusion space **33***h* and in the process space **11***h*.
(65) The effects of the inventive concept are not limited to the above-mentioned effects, and other effects not mentioned above will be clearly understood by those skilled in the art from the above description.
(66) Even though different figures show variations of exemplary embodiments and different embodiments disclose different features from each other, these figures and embodiments are not necessarily intended to be mutually exclusive from each other. Rather, certain features depicted in different figures and/or described above in different embodiments can be combined with other features from other figures/embodiments to result in additional variations of embodiments, when taking the figures and related descriptions of embodiments as a whole into consideration. For example, components and/or features of different embodiments described above can be combined with components and/or features of other embodiments interchangeably or additionally unless the context indicates otherwise.
(67) While embodiments are described above, a person skilled in the art may understand that many modifications and variations are made without departing from the spirit and scope of the inventive concept defined in the following claims. Accordingly, the example embodiments of the inventive concept should be considered in all respects as descriptive/illustrative and not restrictive, with the spirit and scope of the inventive concept being indicated by the appended claims.

Claims

1. A substrate process apparatus comprising: a chamber body providing a process space; and a chamber lid coupled to the chamber body, wherein the chamber lid includes: a focus adapter providing a diffusion space of which width increases in a direction approaching a bottom of the focus adapter; a lid body surrounding the focus adapter; and a coupling adapter supporting the focus adapter, wherein the lid body provides a coupling through hole recessed downwardly from an upper surface of the lid body, wherein the coupling adapter provides a coupling recess recessed downwardly from an upper surface of the coupling adapter, and wherein the coupling recess is spaced apart outwardly from the diffusion space and is located below the coupling through hole.
2. The substrate process apparatus of claim 1, wherein the chamber lid further includes a coupling screw fixing the coupling adapter to the lid body, and wherein the coupling screw is inserted into the coupling through hole and the coupling recess.
3. The substrate process apparatus of claim 1, further comprising a baffle disposed between the

diffusion space and the process space, wherein the diffusion space is connected to the process space through the baffle.
4. The substrate process apparatus of claim 3, wherein the coupling recess is spaced apart upwardly from a lower surface of the coupling adapter, and the coupling recess is not exposed to the process space.
5. The substrate process apparatus of claim 1, wherein the chamber lid further includes a sealing ring between the coupling adapter and the lid body, wherein the sealing ring surrounds the coupling recess in a plan view.
6. The substrate process apparatus of claim 5, wherein the coupling adapter further provides a sealing hole recessed downwardly from an upper surface of the coupling adapter, wherein the sealing hole surrounds the coupling recess in a plan view, and wherein the sealing ring is inserted into the sealing hole.
7. The substrate process apparatus of claim 1, wherein the chamber lid includes a plurality of coupling adapters, and wherein the plurality of coupling adapters are disposed to be spaced apart in a circumferential direction.
8. The substrate process apparatus of claim 7, wherein the chamber lid includes eight coupling adapters.
9. The substrate process apparatus of claim 1, further comprising: an induced coupled plasma (ICP) generator configured to generating plasma; and a chuck configured to support a substrate in the process space, wherein the ICP generator is located on the chamber lid.
10. A substrate process apparatus comprising: a chamber body defining a process space configured such that a substrate is disposed in the process space; and a chamber lid on the chamber body, wherein the chamber lid includes: a focus adapter defining a diffusion space connected to the process space; a lid body surrounding the focus adapter; and a coupling adapter fixing the focus adapter to the lid body, wherein the coupling adapter includes: a support ring; and a plurality of support blocks protruding inwardly from the support ring, wherein the plurality of support blocks are spaced apart from each other in a circumferential direction along the support ring, and wherein the focus adapter is positioned on the plurality of support blocks.
11. The substrate process apparatus of claim 10, wherein the lid body provides a coupling through hole recessed downwardly from an upper surface of the lid body, wherein the support ring provides a coupling recess recessed downwardly from an upper surface of the support ring, and wherein the coupling recess is spaced apart outwardly from the diffusion space and is connected to the coupling through hole.
12. The substrate process apparatus of claim 11, wherein the chamber lid further includes a first sealing ring and a second sealing ring extending in a circumferential direction along the support ring on the support ring, wherein the first sealing ring is located inside the coupling recess, and wherein the second sealing ring is located outside the coupling recess.
13. The substrate process apparatus of claim 12, wherein the support ring includes: a first sealing hole recessed downwardly from an upper surface of the support ring; and a second sealing hole recessed downwardly from the upper surface of the support ring, wherein the first sealing hole is located inside the coupling recess, and wherein the second sealing hole is located outside the coupling recess.
14. The substrate process apparatus of claim 11, wherein the chamber lid further includes a coupling screw fixing the coupling adapter to the lid body, and wherein the coupling screw is inserted into the coupling through hole and the coupling recess.
15. The substrate process apparatus of claim 10, further comprising a plasma generator; and a gas supplier configured to supply gas to the plasma generator, wherein the plasma generator is positioned on the chamber lid and configured to convert the gas supplied from the gas supplier into plasma.
16. The substrate process apparatus of claim 10, wherein the lid body includes quartz.

17. The substrate process apparatus of claim 10, wherein the support ring and the plurality of support blocks include aluminum (Al).
18. A substrate process method comprising: placing a substrate in a substrate process apparatus; supplying a gas to the substrate process apparatus; generating plasma using the gas supplied to the substrate process apparatus; and processing the substrate with the plasma, wherein the substrate process apparatus includes: a chamber body providing a process space in which the substrate is disposed; and a chamber lid on the chamber body, wherein the chamber lid includes: a focus adapter providing a diffusion space in which the plasma is diffused; a lid body surrounding the focus adapter; and a coupling adapter supporting the focus adapter, wherein the lid body provides a coupling through hole connected to an upper surface of the lid body, and wherein the coupling adapter provides a coupling recess positioned below the coupling through hole.
19. The substrate process method of claim 18, wherein the substrate process apparatus further includes a baffle between the diffusion space and the process space, wherein the diffusion space is connected to the process space through the baffle, and wherein the processing of the substrate with the plasma includes moving the plasma from the diffusion space to the process space through the baffle.
20. The substrate process method of claim 18, wherein the processing of the substrate using the plasma includes removing a photoresist (PR) from the substrate by the plasma.